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Passive vs. Active Battery Balancing: How to Choose the Right Strategy for Mobile Robots

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A mobile robot can have the right battery capacity on paper and still struggle to deliver the expected runtime.

Sometimes, the issue is not simply how much energy the battery stores. It is how effectively the battery pack can use that energy across all its cells.

This is where cell balancing comes in.

For AGV, AMR and UGV manufacturers, passive and active balancing are two approaches worth understanding. But is active balancing always the better choice? And when does the additional cost and complexity actually make sense?

The answer depends on how the robot operates, how it charges and what the battery system needs to achieve over its service life.

1. Why Does Cell Balancing Matter?

In a battery pack made up of cells connected in series, each cell contributes to the overall pack voltage. Ideally, the cells should have closely matched characteristics and operate under similar conditions.

In practice, small differences can develop due to manufacturing variation, temperature differences, charging and discharging patterns, and ageing.

Over time, these differences can become more noticeable.

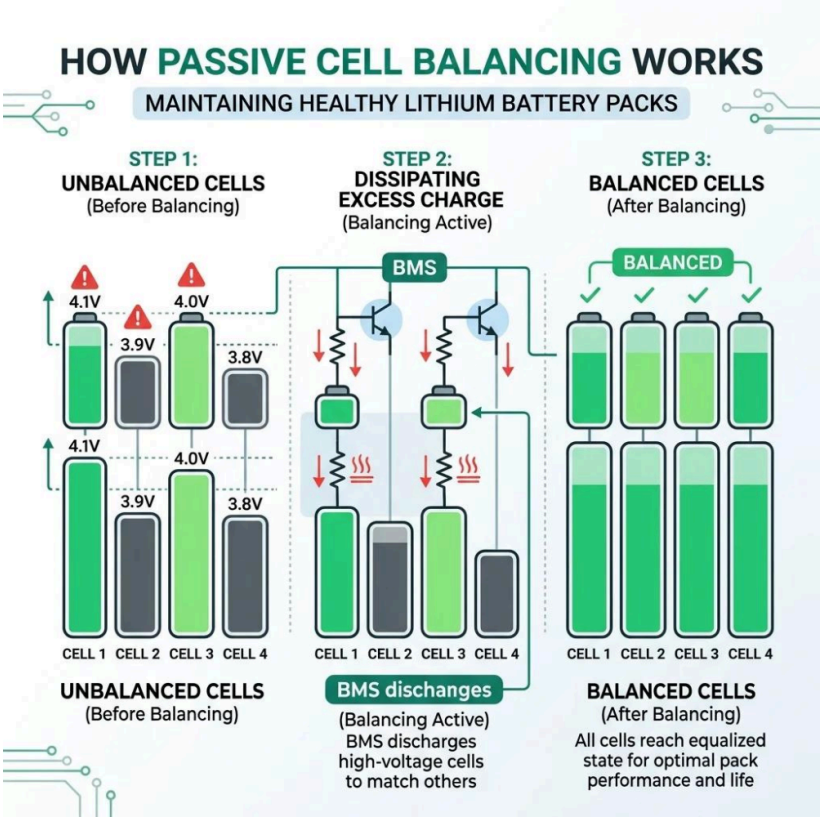
During charging, one cell may reach its upper voltage limit before the others. During discharging, another may reach its lower voltage limit first.

Because the battery management system (BMS) must protect every cell, the first cell to reach a protection threshold can limit the charging or discharging of the entire pack.

The result may be less usable energy than expected.

Cell balancing helps manage differences in cell state of charge (SOC). However, it is important to understand that balancing does not restore lost cell capacity or repair a damaged cell.

The goal is not simply to make cell voltages look similar. It is to help the battery pack operate within its intended limits while making effective use of the energy available.



2. Passive vs. Active Balancing: What Is the Difference?

The two methods address cell imbalance in different ways.

Passive Balancing: Dissipating Excess Energy

Passive balancing uses resistors to discharge cells with a higher state of charge. The excess energy is converted into heat rather than being transferred to other cells.

Its main advantages are simplicity, relatively low cost and a well-established design approach.

For many industrial battery packs, passive balancing is a practical solution, particularly when cell matching is good, imbalance is limited and the charging schedule allows sufficient time for balancing.

There are trade-offs, however.

Balancing currents are often relatively low, and the energy dissipated as heat is not recovered. The available balancing time and thermal design can therefore affect how well the system manages cell differences.

Active Balancing: Transferring Energy Between Cells

Active balancing transfers energy from cells with a higher state of charge to cells with a lower state of charge through circuits using components such as capacitors, inductors or DC-DC converters.

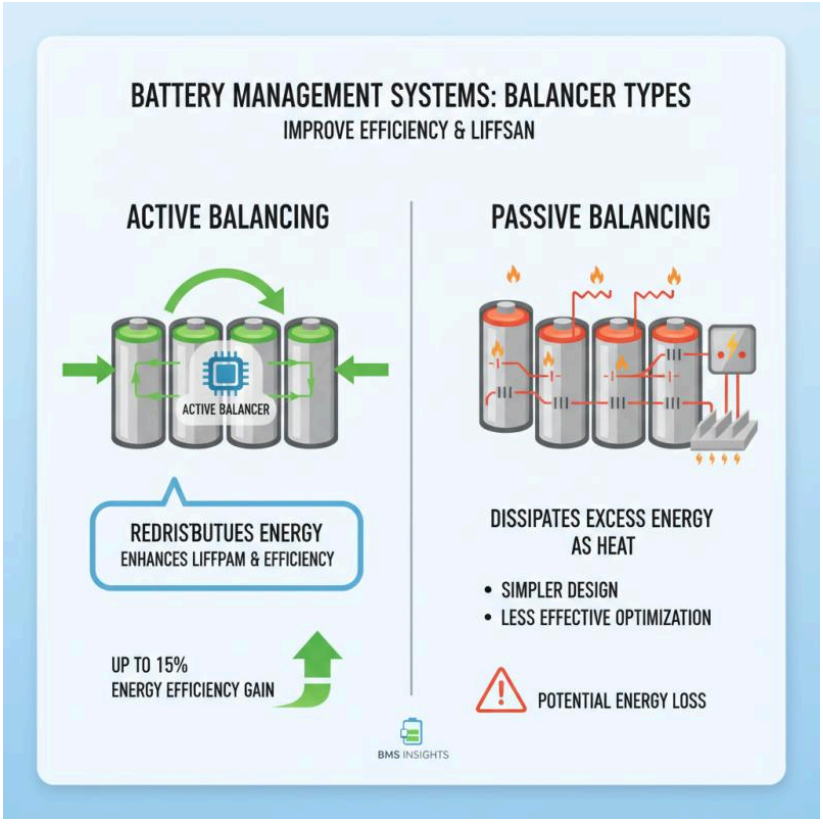
Instead of dissipating all the excess energy, the system redistributes part of it within the battery pack.

Depending on the design, active balancing can support balancing during charging and discharging, reduce balancing-related energy losses and improve the utilisation of available cell capacity.

The trade-off is greater circuit and control complexity, additional components, and potentially higher cost.

Its actual performance depends on the balancing current, transfer efficiency, control strategy, operating conditions and the degree of cell imbalance.

Active balancing offers more options for managing energy. It does not automatically make every battery system more reliable or cost-effective.



3. Which Approach Makes More Sense for AGV, AMR and UGV Applications?

The choice should start with the robot's operating profile, not with the BMS feature list.

AGV: Predictable Routes and Charging Schedules

Many AGVs operate on relatively predictable routes and return to a charging station according to a planned schedule.

If the battery has well-matched cells, the imbalance remains limited and charging periods provide enough time for the BMS to balance the cells, passive balancing may be entirely sufficient.

The key questions are whether the balancing current is adequate and whether the available balancing time matches the battery's actual requirements.

AMR: Frequent Operation and Short Charging Windows

Some AMRs operate for long periods, with only short opportunities to recharge between tasks.

In these applications, engineers may need to pay closer attention to balancing time, cell consistency and usable capacity.

Active balancing may be worth evaluating when the battery pack has meaningful cell imbalance and the charging schedule leaves little time for conventional balancing.

However, active balancing cannot compensate for an undersized battery, excessive energy consumption or a cell that has significantly degraded.

Before selecting it, engineers should establish whether cell imbalance is actually limiting the system's performance.

UGV: Variable Loads and Challenging Environments

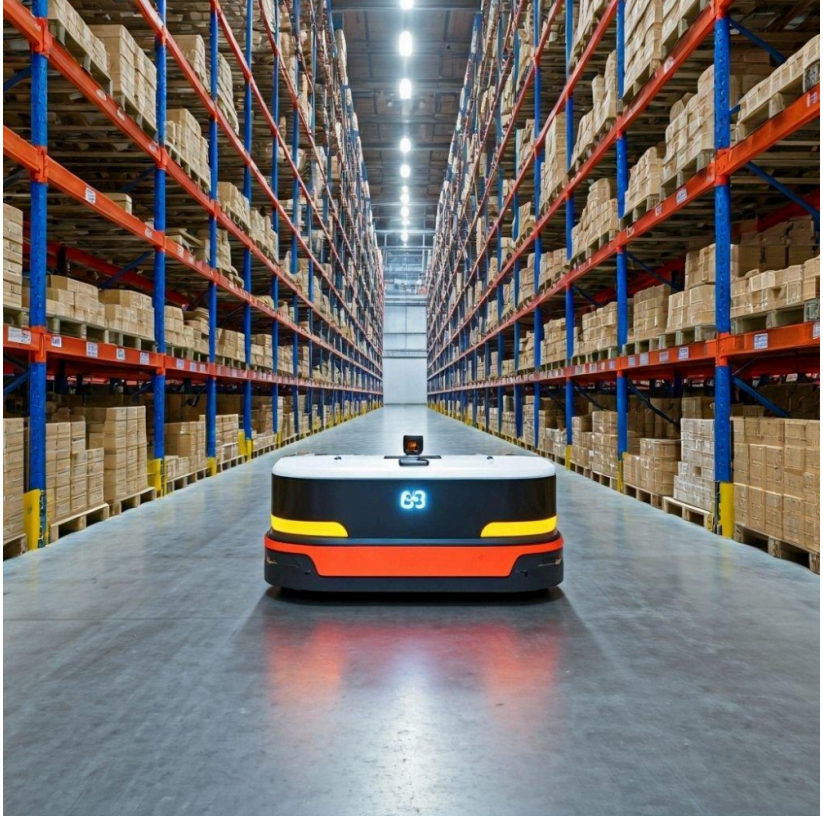
UGVs may operate outdoors, carry changing payloads or encounter wide variations in ambient temperature.

Here, cell balancing is only one part of the battery management strategy.

Temperature monitoring, cell matching, thermal management, discharge limits and protection settings can be equally important. An active balancing circuit may offer benefits in some designs, but it should be assessed alongside these other requirements.

The important point is that **AGV, AMR and UGV are application categories, not automatic rules for choosing a balancing method.**

Two robots in the same category may need very different battery architectures.



4. Five Questions to Ask Before Choosing a Balancing Strategy

Before specifying passive or active balancing, I would start with five practical questions.

1. How long does the robot operate between charging opportunities?

A battery that has several hours to charge and balance has different requirements from one that receives only brief charging windows throughout the day.

2. How much cell imbalance needs to be corrected?

A small imbalance in a well-matched battery pack may not justify a more complex solution. Larger imbalances or limited balancing time may warrant further investigation.

3. What balancing current does the system actually need?

The label “active balancing” tells you very little by itself. The balancing current, transfer efficiency and control strategy determine how quickly and effectively the system can redistribute energy.

For passive balancing, engineers should also consider the heat generated and the time required to correct the expected imbalance.

4. What are the battery chemistry and operating conditions?

Cell chemistry, voltage limits, temperature distribution, ageing and charging behaviour all influence the design.

Balancing thresholds and control strategies should be appropriate for the selected cells rather than copied from another battery application.

5. What is the real lifetime cost?

A more sophisticated BMS may improve energy utilisation in a particular application, but the additional electronics and integration work also have a cost.

The decision should consider usable capacity, charging requirements, maintenance, downtime and system reliability over the intended service life.

The objective is not to choose the most advanced feature. It is to choose a solution that delivers measurable value in the actual application.

5. A Simple Engineering Comparison

Consideration	Passive Balancing	Active Balancing
Energy management	Excess energy is dissipated as heat	Energy is transferred between cells
Circuit complexity	Generally lower	Generally higher
Cost	Generally lower	Generally higher
Balancing speed	Limited by balancing current and thermal constraints	Depends on transfer current, topology and control
Charging and discharging	Commonly performed during charging	Some designs support balancing during both charging and discharging
Best reason to choose it	Simplicity and cost-effectiveness meet the application needs	Energy redistribution or higher balancing capability offers a meaningful benefit

This comparison describes general design tendencies, not guaranteed performance. Actual results depend on the BMS architecture, cell characteristics and operating conditions.

6. When Should You Investigate Active Balancing?

Active balancing deserves a closer look when several conditions occur together:

- The battery pack has a meaningful and persistent cell imbalance.
- The operating schedule leaves limited time for balancing.
- Improving the utilisation of available cell capacity is an important design objective.
- The expected benefit justifies the additional cost and system complexity.

If the battery is consistently underperforming, however, balancing should not be the first assumption.

Check the cell voltages under relevant operating conditions, temperature distribution, cell capacity and internal resistance differences, charging behaviour, and BMS protection records.

A single voltage reading may not tell the whole story, particularly when the cells are under load.

The right diagnosis should come before a change in BMS architecture.

My Opinion

For mobile robot manufacturers, battery balancing is not simply a choice between a basic and an advanced BMS.

It is a decision about how the battery system should behave throughout the robot's working life.

Passive balancing may be the right answer when the system is well matched, charging opportunities are sufficient and simplicity is a priority.

Active balancing may be worth the investment when the application has specific energy utilisation or balancing requirements that justify the additional complexity.

Neither approach replaces good cell selection, sound thermal design or appropriate battery protection.

The best balancing strategy is the one that matches the robot's real operating conditions — not the one with the longest feature list.

When designing your next AGV, AMR or UGV, how do you evaluate cell balancing alongside battery capacity, charging time and BMS protection?

#BatteryManagementSystem #BatteryTechnology #MobileRobotics #AMR #AGV #UGV #LithiumIonBatteries

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